

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$$

Direction table *u* ∈ {x, y, z} = axis perp. to boundary

Inc dir	Inc	Refl	Trans
+û	$e^{-jk_1 u}$	$e^{+jk_1 u}$	$e^{-jk_2 u}$
-û	$e^{+jk_1 u}$	$e^{-jk_1 u}$	$e^{+jk_2 u}$

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m) (incident wave, Med 1)

$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m) (reflected wave, Med 1)

$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m) (transmitted wave, Med 2)

(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)

Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T SUMMARY

Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.

	Normal ⊥ pol (θ _i = 0)	pol
Γ wave	refl. $\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$
τ wave	trans. $\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$
T pwr	trans. $ \tau ^2$	$ \tau_{\perp} ^2$
R + T total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$

Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; *nonmag.* ⇒ $\eta_2 / \eta_1 = n_1 / n_2$.

Intrinsic impedance η_i *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says “ferromagnetic”/iron/ferrite/etc.

Low-loss correction η_c $\sigma/(\omega\epsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

For Γ/τ just use $\eta_0/\sqrt{\epsilon_r}$. The j term is for |η_c|, θ_η. See LOSSY MEDIA — 5 CASES for all regimes.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η-ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: %P_r + %P_t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs 1 → 2 → 3 → 4*

$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT

Beam offset through *N* slabs thickness *t_i*, refracted angle inside slab *i*

Slab-indexed θ_i = angle inside slab *i*; *t_i* = slab thickness (*m*)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (*t_i*).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

Type	$\sigma/\omega\epsilon$	η_c, α, β
Lossless ($\sigma=0$)	0	$\eta = \eta_0/\sqrt{\epsilon_r}$ exact $\alpha=0, \beta = \omega\sqrt{\epsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right)$ $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega\sqrt{\epsilon_r}}{c}$
Quasi-cond.	10^{-2} – 10^2	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$ see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f \mu/\sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\epsilon_r} \eta_0$

For Γ/τ: drop j correction; use $\eta_0/\sqrt{\epsilon_r}$. Magnetic: keep μ_r .

Low-loss quick params $\sigma/(\omega\epsilon) \ll 1, \mu_r = 1$

$$\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}} \text{ (Np/m)}, \beta \approx \frac{\omega\sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}$$

Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$

$$\alpha = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X-1}, \beta = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X+1}$$
$$\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'}$$

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$

TM mode $H_z = 0, E_z \neq 0$

Dimensions *a* × *b* with *a* > *b* along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x form (TE) *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on x = mπ/a; coef. on y = nπ/b.

Identify TE_{*m**n*} vs TM_{*m**n*} *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. “a TE wave” → TE).

2. If *m*=0 or *n*=0 in the mode label ⇒ must be TE (TM forbids 0).

3. Source field *H_z*(*x, y*) given ⇒ TE. Source field *E_z*(*x, y*) given ⇒ TM.

TE allows at least one of *m, n* ≥ 1 (other can be 0). TM needs both *m, n* ≥ 1. That's why TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

All fields carry an $e^{-j\beta z}$ factor (omitted below). $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$.

TE mode *generator: $\hat{H}_z$*

$$\hat{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$
$$\hat{E}_x = -\frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$
$$\hat{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$
$$\hat{E}_z = 0, \hat{H}_x = -\hat{E}_y/Z_{TE}, \hat{H}_y = \hat{E}_x/Z_{TE}$$

TM mode *generator: $\hat{E}_z$*

$$\hat{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$
$$\hat{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$
$$\hat{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$
$$\hat{H}_z = 0, \hat{H}_x = -\hat{E}_y/Z_{TM}, \hat{H}_y = \hat{E}_x/Z_{TM}$$

TEM (plane wave) for reference: $\hat{E}_x = E_0 e^{-j\beta z}, \hat{H}_y = \hat{E}_x/\eta, f_c$ N/A, $k = \omega\sqrt{\mu\epsilon}$.

CUTOFF FREQUENCY

f_{mn} (a.k.a. *f_c* for that mode) *m, n integers* ≥ 0 (not both zero)

$$f_c \equiv f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

Common cutoffs (*a* > *b*)

Mode	<i>f_{mn}</i>
<i>u_{p0}</i>	<i>c</i> /√ε _r (hollow: <i>u_{p0}</i> = <i>c</i>)
TE ₁₀	<i>f</i> ₁₀ = <i>u_{p0}</i> /(2 <i>a</i>) (dominant)
TE ₀₁	<i>f</i> ₀₁ = <i>u_{p0}</i> /(2 <i>b</i>)
TE ₂₀	<i>f</i> ₂₀ = <i>u_{p0}</i> / <i>a</i> = 2 <i>f</i> ₁₀
TE ₁₁ , TM ₁₁	<i>f</i> ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires *f* > *f_{mn}*. In *Z_{TE}*/*Z_{TM}* and *u_g*, *f_c* = *f_{mn}* of the excited mode.

ε_r, FROM DISPERSION

Given ω, β, mode (*m, n*) *result is unitless*

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] \text{ (unitless)}$$

Inputs: ω (rad/s), β (rad/m), *a, b* (m), *c* = 3 × 10⁸ (m/s), *m, n* (unitless).

WAVE IMPEDANCE & *H_y*

$$Z_{TE} = \frac{\eta}{\sqrt{1-(f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{TM} = \eta \sqrt{1-(f_c/f)^2}$$
$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}$$

Z_{TE} > η always (denominator < 1); *Z_{TM}* < η.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$
$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \text{ (m/s)} \Rightarrow u_p u_g = u_{p0}^2 \text{ (m}^2\text{/s}^2\text{)}$$

t = *L*/*u_g* (s) travel time through guide of length *L*

Higher modes have lower *u_g* ⇒ they arrive later. Group delay spread = sum over excited modes.

BOUNDARY

η_1, η_2 — imp. (Ω)

Γ, τ, *R, T* — coeffs (unitless)

τ = 1 + Γ; *R* = |Γ|²

$k_i = \omega\sqrt{\epsilon_{r,i}}/c$ (rad/m)

α₂ (Np/m), β₂ (rad/m): lossy

z=0 — boundary plane (m)

η₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_i, θ_t — angles from normal

n_i = √ε_{*r, i*} — index of refr.

t_i — slab thickness (m)

d — lateral disp. (m)

plane of inc. = plane of $\hat{k}$ +normal

⊥: **E** ⊥ this plane

||: **E** || this plane

WAVEGUIDE

a, b — wide, narrow dim. (m); *a* > *b*

m, n — mode indices (int.)

f_{mn} — cutoff (Hz)

β — prop. const (rad/m)

Z_{TE}/*Z_{TM}* — wave imp. (Ω)

E_x, H_y — field comps (V/m, A/m)

η = η₀/√ε_r

VELOCITIES

u_{p0} = *c*/√ε_r — bulk phase vel.

$$u_p = u_{p0} / \sqrt{1 - (f_c/f)^2}$$
$$u_g = u_{p0} \sqrt{1 - (f_c/f)^2}$$

u_p *u_g* = *u_{p0}*²

t = *L*/*u_g* — travel time

c = 3 × 10⁸ m/s

POWER & UNITS

%*P_r* = 100|Γ|²

%*P_t* = 100|τ|²η₁/η₂

%*P_r* + %*P_t* = 100

σ (S/m), ε (F/m), μ (H/m)

ε₀ = 8.85 × 10⁻¹² F/m

μ₀ = 4π × 10⁻⁷ H/m

1/(36π) × 10⁻⁹ ≈ ε₀